

Commercial Pilot (ASEL) Oral Exam Question Bank

A knowledge-focused study resource for the Commercial Pilot Airplane Single-Engine Land checkride oral exam. Organized by topic area, with a complete model answer for every question.

Two tips before you start studying: - **Answer the question asked, then stop.** Volunteering extra information beyond what was asked often invites a harder follow-up question that wasn't coming. If the examiner wants more detail, they'll ask for it. - **Speak in your own words, not acronyms.** An examiner can tell when an answer is recited from a mnemonic rather than understood — be ready to explain the substance behind any acronym you use.

1. Certificates, Privileges & Limitations

1. What are the eligibility requirements to obtain a commercial pilot certificate? **A:** At least 18 years old; able to read, speak, write, and understand English; hold at least a private pilot certificate (or meet the requirements concurrently); pass the commercial aeronautical knowledge test and the practical test; and meet the aeronautical experience requirements of 61.129 — 250 hours of flight time for an airplane single-engine rating.
2. What's the aeronautical experience breakdown behind that 250-hour number? **A:** At minimum: 100 hours in powered aircraft (50 in airplanes), 100 hours of PIC time (50 in airplanes, 50 cross-country of which 10 in airplanes), 20 hours of training on the commercial maneuvers including 10 hours of instrument training and 10 hours in a complex, turbine-powered, or technically advanced airplane, plus specific day and night cross-country training flights, and 10 hours of solo (or PIC-with-instructor-aboard) time including a 300nm three-point solo cross-country and 5 hours of night VFR work.
3. What medical certificate is required to exercise commercial pilot privileges? **A:** At least a second-class medical certificate to be compensated for acting as PIC; a third-class medical (or BasicMed, where its limitations allow) is sufficient if you're exercising private pilot privileges instead, since a higher class of medical isn't required unless you're actually being compensated for that specific flight.
4. What are the basic privileges and limitations of a commercial pilot certificate? **A:** You may act as PIC for compensation or hire, carrying passengers or property, subject to the operating rules for the kind of flight being conducted (generally Part 91 for most commercial single-engine operations, since Parts 121/125/135 govern scheduled or larger-scale commercial carriage). You're still restricted from operations that require an air carrier certificate unless you or your employer holds one.
5. What is the difference between common carriage and private carriage, and why does it matter to a commercial pilot? **A:** Common carriage exists when an operator "holds out" — advertises, or otherwise signals a general willingness — to transport persons or property from place to place for compensation to anyone who wants it. That triggers the requirement for an air carrier certificate under Part 121 or 135. Private carriage for hire is carriage for one or a limited number of selected customers, generally on a long-term or specifically negotiated basis, without holding out to the general public — that can be conducted under Part 91. A commercial pilot can fly private carriage on their own certificate; common carriage requires

the operator (often not the individual pilot) to hold economic authority as an air carrier.

6. A client wants to pay you to fly them and a package from one airport, to a second airport to run an errand, and back. Is this legal on a commercial certificate alone? **A:** Not as described — transporting a person and property from place to place for compensation is exactly what common carriage means, and doing it under Part 91 alone (without an air carrier certificate) isn't legal. It can become legal by restructuring the flight so it isn't point-to-point paid transportation for a third party — for example, being contracted directly by an organization for a specific mission where the aircraft returns to its origin and the flying is incidental to that business relationship, rather than the passenger paying for transportation between two points.

7. If you're operating a compensated flight and a passenger needs a bathroom or food stop, does landing at an unplanned airport turn the flight into illegal point-to-point transportation? **A:** No — the FAA has specifically addressed this. Stopping to accommodate a passenger's bodily needs (food, restroom) doesn't constitute providing transportation to that stop as a destination, so it doesn't create a common carriage problem on its own. The distinction is whether the airport itself was a paid destination versus an incidental stop for a legitimate need during an otherwise properly structured flight.

8. What commercial operations can a pilot conduct under Part 91, without an air carrier certificate? **A:** 14 CFR 119.1(e) lists the specific categories excepted from the Part 119 air carrier certificate requirement, for any airplane under 20 passenger seats or 6,000 lb payload — which covers essentially all ASEL commercial work: student instruction; nonstop commercial air tours conducted under a Letter of Authorization (LOA) from the FSDO, beginning and ending at the same airport within a 25-statute-mile radius; ferry or training flights; aerial work — crop dusting/seeding/spraying, banner towing, aerial photography or survey, firefighting, and powerline/pipeline patrol; and nonstop flights within 25 statute miles of the takeoff airport carrying parachute jumpers. Critically, this exception applies *regardless of whether common carriage is involved* — meaning these specific operations can be openly advertised and held out to the general public and still legally operate under Part 91 alone, unlike most other commercial-for-hire flying, where holding out is exactly what triggers the need for an air carrier certificate under Part 121 or 135.

9. What does "pro rata" sharing of expenses mean, and how is it different from being paid to fly? **A:** Pro rata means splitting only the actual operating costs of a specific flight — fuel, oil, aircraft rental, landing/tie-down fees — evenly among the occupants who could act as PIC and the passengers, with the pilot never receiving more than an equal share. It's not compensation for piloting; a commercial certificate isn't required for it, and a private pilot can do it too, as long as there was a genuine shared reason to make the flight (not one arranged solely to generate income for the pilot).

10. What is "holding out," and what are some real-world examples that cross the line? **A:** Advertising or otherwise signaling a general willingness to fly anyone who wants it, for compensation, from place to place — signs, advertising, a pattern of accepting requests from the general public, or even word-of-mouth reputation for being willing to serve all customers can qualify. A private pilot posting on social media asking if anyone wants to split gas on a flight is a commonly cited example of holding out, since it's soliciting the general public rather than sharing costs with someone who already had a reason to go. This is the general rule for point-to-point paid transportation; the 119.1(e) categories in the previous question (aerial work, LOA air tours, and the others) are a specific, named exception to it, not a contradiction of it.

11. What documents must you have in your possession to act as pilot-in-command? **A:** Government-issued photo ID, pilot certificate, and current medical certificate appropriate to the privileges being exercised (or

BasicMed documentation, if applicable and if its limitations permit the flight).

12. What's required for you to remain current to carry passengers? **A:** Three takeoffs and three landings to a full stop in the same category/class (and type, if required) of aircraft within the preceding 90 days. Night passenger currency requires those landings be to a full stop, made between one hour after sunset and one hour before sunrise; meeting the night requirement also satisfies day currency.

13. How do you keep your certificate current to act as PIC at all, separate from passenger-carrying currency? **A:** A flight review — 1 hour ground, 1 hour flight, with an authorized instructor — every 24 calendar months. Completing a new certificate or rating, a proficiency check, or a phase of the FAA WINGS program can substitute for it.

2. Aircraft Leasing & Operational Control

14. What is the difference between a wet lease and a dry lease? **A:** A wet lease is any leasing arrangement where a person agrees to provide an entire aircraft and at least one crewmember. A dry lease is leasing an aircraft without a crew. In a dry lease, the lessee normally exercises operational control of the aircraft; in a wet lease, the lessor normally does — though the real determination depends on who actually exercises authority over the flight, not just what the lease document calls itself.

15. What is "operational control," and what three areas does it cover? **A:** The exercise of authority over initiating, conducting, or terminating a flight. It covers three areas: Aircrew (ensuring crewmembers are trained, qualified, and in compliance with duty/rest rules, and designating a PIC for each flight), Aircraft (ensuring it's airworthy and compliant with applicable regulations), and Flight Management (weather minimums, loading/CG, icing, fuel requirements, and the monetary/logistical issues tied to the aircrew and aircraft).

16. If you dry lease an aircraft, do you need an FAA operator's certificate? **A:** Not as long as you don't carry persons or property for compensation or hire — dry leasing without compensation puts you in essentially the same position as any private operator under Part 91. The moment compensation for carrying persons or property enters the picture, the operational-control questions above become the determining factor for whether an air carrier certificate is actually required.

17. What is a "sham dry lease," and why should a pilot or operator be alert to it? **A:** An arrangement structured on paper as a dry lease — implying the lessee has operational control — while in practice the leasing company still services and flies the aircraft, retaining real operational control. It's a way some operators evade air carrier certification requirements while appearing compliant. A pilot or lessee who unknowingly accepts what amounts to disguised charter service loses the protection of the higher safety standards an actual air carrier certificate requires, and may be held legally responsible for regulatory violations they didn't realize they'd taken on.

18. What questions help determine who actually holds operational control of a flight, regardless of what the lease is named? **A:** Who decides to assign crew and aircraft, and to accept, initiate, conduct, and terminate flights? Who do the pilots work for? Who maintains the aircraft, and where? Who ensures the flight, aircraft, and crew comply with regulations before departure? Who decides when maintenance happens and pays for

it? Who determines weather/fuel requirements and pays for fuel? Who pays for airport fees, parking, and related costs? Answering "yes, I'm responsible" to any of these means holding at least some operational control — and the responsibilities, and liability, that come with it.

3. Aircraft Documents & Airworthiness

19. What documents must be in the aircraft for it to be legal to fly? **A:** Airworthiness certificate, Registration, Radio station license (international operations only), Operating limitations/handbook, and Weight and balance data — plus placards, the compass deviation card, and any supplements for equipment installed after manufacture. (Mnemonic: ARROW.)

20. What inspections does the aircraft require, and how often? **A:** Annual — every 12 calendar months. 100-hour — every 100 hours of operation, required only if the aircraft is used for hire or flight instruction for hire (an annual satisfies the 100-hour requirement, not the reverse). ADs — complied with on whatever schedule each specifies. VOR check — every 30 days, only for IFR flight using VOR navigation. Transponder — every 24 calendar months. ELT — every 12 calendar months, and also after 1 cumulative hour of use or 50% battery life expended. Static system/altimeter/pitot — every 24 calendar months, for IFR flight only.

21. What is a KOEL, and how is it different from an MEL? **A:** A KOEL (Kinds of Operation Equipment List), found in the POH limitations section, lists the equipment required for specific kinds of operation — day VFR, night VFR, IFR — and applies to virtually every GA aircraft. An MEL (Minimum Equipment List) is an FAA-approved list of what may legally be inoperative for dispatch, created by an operator and typically serialized to a specific tail number rather than just an aircraft type — most single-engine trainers don't have one, so 91.213(d) governs instead.

22. What would you do if you discovered inoperative equipment before flight? **A:** Determine whether the item is required by the type certificate, an AD, 91.205, or the kind of operation being flown. If it's not required by any of those, it can be placarded "Inoperative" and deactivated under 91.213(d) — but only once a certificated mechanic determines it's safe to fly without it. If it is required equipment, the aircraft isn't airworthy until it's repaired, replaced, or an approved MEL covers it.

23. Can a pilot personally disable or placard a piece of inoperative equipment themselves — for example, an attitude indicator? **A:** Generally no. A pilot's authority is limited to the specific items listed in Part 43 Appendix A(c) — oil and filter changes, bulb/fuse replacement, a handful of similar preventive maintenance tasks — and disabling an instrument (an attitude indicator, for example, tied into a shared vacuum system) isn't on that list. That determination and the physical deactivation require a certificated mechanic, even when the fix sounds simple, like pulling a circuit breaker — a breaker can be shared with other equipment in ways that aren't obvious from the panel alone, and pulling one without documented authorization to do so isn't legal.

24. What is a Supplemental Type Certificate (STC), and what if you want to install something with no STC available? **A:** An STC is FAA approval for a specific modification or piece of equipment to be added to an already-certificated aircraft, obtained by the company that developed the modification after it goes through FAA safety testing. If no STC exists for what you want to install, the path is a field approval — working with

an A&P and the local FSDO to get the specific installation approved for that aircraft, since there's no blanket path around either an STC or a field approval to modify a certificated airplane.

4. Complex, High-Performance, and Technically Advanced Aircraft

25. What makes an airplane "complex" under 61.31, and why does the commercial certificate require training in one? **A:** A complex airplane has retractable landing gear, flaps, and a controllable-pitch propeller (or is a seaplane satisfying an equivalent standard). 61.129 requires at least 10 hours of training in a complex, turbine-powered, or technically advanced airplane specifically so the additional systems knowledge and workload of a higher-performance airplane are addressed before commercial certification, not left for later.

26. What is a technically advanced airplane (TAA), and what must it have to qualify? **A:** An airplane equipped with an electronic Primary Flight Display (at minimum an airspeed indicator, turn coordinator, attitude indicator, heading indicator, altimeter, and vertical speed indicator), an electronic Multifunction Display with a GPS-based moving map, and a two-axis autopilot integrated with the navigation and heading guidance — all continuously visible in flight. It can satisfy the same 61.129 training requirement as a complex airplane.

27. What's the risk profile difference in operating a retractable-gear airplane versus a fixed-gear one? **A:** The primary added risk is gear-up landing — a pilot must positively verify gear position and follow a disciplined pre-landing checklist rather than relying on habit, since the airplane will often fly a perfectly normal approach and landing right up until touchdown regardless of gear position. Systems risk also increases: a hydraulic or electrical failure can leave the gear in an unknown or partially-extended state, requiring an emergency extension procedure specific to that aircraft.

28. How does a constant-speed propeller differ from a fixed-pitch propeller in principle, and why does that matter operationally? **A:** A constant-speed prop uses a governor to vary blade pitch automatically, holding a selected RPM across a range of airspeeds and power settings — separating power (manifold pressure) from RPM as two independently controlled variables, unlike a fixed-pitch prop where they're linked. Operationally this means power and RPM changes have to be sequenced correctly (generally: increase RPM before increasing manifold pressure, decrease manifold pressure before decreasing RPM) to avoid unnecessary stress on the engine.

29. What additional endorsement does a pilot need before acting as PIC of a complex, high-performance, pressurized, or tailwheel airplane? **A:** Each requires its own specific ground and flight training plus a logbook endorsement from an authorized instructor under 61.31 — complex, high-performance (defined as more than 200 horsepower), pressurized aircraft capable of operating at high altitude, and tailwheel. These are separate from, and in addition to, any rating already on the certificate — an instructor has to specifically verify the training and proficiency for each one before signing it off, and a pilot can't legally act as PIC of that category of airplane without it.

5. Weather

30. How can you obtain a weather briefing before flight, and what types are there? **A:** 1800wxbrief.com, the ForeFlight/Garmin Pilot app, AviationWeather.gov, or calling Flight Service. A Standard briefing is complete, used for the first briefing of the day or after 6+ hours since the last one. An Abbreviated briefing updates a previous one or covers specific information only. An Outlook briefing is used when departure is 6+ hours away, for planning only, and must be followed by a standard or abbreviated briefing closer to departure.

31. What's a METAR, a TAF, and a PIREP? **A:** A METAR is a routine, actual-conditions surface observation issued hourly (or as a special/SPECI when conditions change significantly). A TAF is a forecast issued roughly every 6 hours, valid 24–30 hours. A PIREP is a pilot report of actual in-flight conditions — the only "ground truth" source of current en route weather, since forecasts are inferential and a PIREP is an eyewitness account.

32. What are AIRMETs and SIGMETs, and how long is each valid? **A:** SIGMETs cover hazards significant to all aircraft — severe icing, severe/extreme turbulence, dust/sandstorms below 3 SM visibility, volcanic ash — valid up to 4 hours (6 for hurricanes). Convective SIGMETs cover severe thunderstorm activity, valid 2 hours. AIRMETs cover less severe hazards of particular concern to lighter aircraft, issued every 6 hours: Sierra (IFR/mountain obscuration), Tango (moderate turbulence, sustained winds ≥ 30 kt, low-level wind shear), Zulu (moderate icing and freezing levels).

33. What are the types of structural icing, and what two conditions does icing require? **A:** Rime ice is rough, milky, and forms quickly from small supercooled droplets, typical in stratiform clouds. Clear ice is smooth, hard, transparent, and the most dangerous and hardest to remove, from larger supercooled droplets that spread before freezing. Mixed ice is a combination of both. Structural icing always requires visible moisture plus a temperature at or below freezing at the airframe surface.

34. What are the VFR weather minimums? **A:** Class B: 3 SM, clear of clouds. Class C/D/E below 10,000 ft MSL: 3 SM, 500/1,000/2,000 ft from clouds. Class E at or above 10,000 ft MSL: 5 SM, 1,000/1,000/1 SM. Class G at or below 1,200 ft AGL: 1 SM day (clear of clouds) / 3 SM night (500/1,000/2,000). Class G above 1,200 ft AGL below 10,000 ft MSL: 1 SM day / 3 SM night, 500/1,000/2,000. Class G above 1,200 ft AGL at/above 10,000 ft MSL: 5 SM, 1,000/1,000/1 SM.

35. Describe the different fronts and what weather each typically brings. **A:** A warm front rides warm air up over retreating cooler air on a shallow slope, bringing widespread stratiform clouds and steady precipitation well ahead of the surface front, moving slowly. A cold front wedges cold air under warm air abruptly on a steep slope, bringing a narrow band of cumuliform clouds, showery or heavy precipitation, turbulence, and sometimes thunderstorms, moving fast with rapid clearing behind it. An occluded front forms when a faster cold front catches a slower warm front — expect warm-front weather followed by cold-front weather in sequence. A stationary front has two air masses meeting with little relative motion, and its weather can linger for days.

36. What three ingredients does a thunderstorm need, and what should a commercial pilot's personal minimums around convective weather look like? **A:** Sufficient moisture, an unstable air mass, and a lifting force. Because a commercial pilot is often flying for a client or a schedule, the external-pressure risk around thunderstorms is higher than in private flying — personal minimums should be set and communicated to clients or employers in advance, precisely so a weather decision doesn't get made under pressure in the moment.

37. Does density altitude only matter for takeoff and landing distance? **A:** No — it also affects climb performance after liftoff. A runway can be more than adequate on a hot, high-elevation day while the airplane still can't outclimb rising terrain ahead, especially relevant for a commercial pilot who may be flying heavier loads (passengers, cargo, camera or survey equipment) than typical private training flights.

6. Airspace

38. Name the six classes of airspace and how their requirements differ. **A:** Class A — 18,000 ft MSL to FL600, IFR only. Class B — requires an explicit ATC clearance, two-way radio, Mode C transponder, and ADS-B Out. Class C — two-way radio contact established before entry, Mode C transponder required. Class D — two-way radio contact required, around towered airports without approach control. Class E — controlled airspace not otherwise classified. Class G — uncontrolled, the absence of B/C/D/E markings.

39. What are the transponder emergency codes? **A:** 1200 is standard VFR (no ATC services). 7500 is hijacking, 7600 is lost communications, 7700 is a general emergency.

40. What are the basic right-of-way rules? **A:** An aircraft in distress has right of way over all other traffic. When converging (other than head-on) at the same altitude, the aircraft to the other's right has the right of way — except a balloon, glider, or towing/refueling aircraft has right of way over powered aircraft. Head-on, both alter course right. Overtaking, the aircraft being overtaken has the right of way and the overtaking aircraft passes to the right. On landing or final approach, the aircraft at the lowest altitude has right of way, but may not use that to cut off an aircraft already established closer to the runway.

41. What is a TFR, and what special use airspace types should you know? **A:** A Temporary Flight Restriction is issued by NOTAM for events like VIP movement, wildfires, or disasters and is typically mandatory to avoid. Prohibited Areas bar flight entirely; Restricted Areas prohibit entry when active without permission (hazardous activity like gunnery); Warning Areas are the same hazards over international waters; MOAs permit VFR flight but warrant caution during military training activity; Alert Areas flag high training/unusual-activity volume with no special clearance required; National Security Areas are voluntary-avoidance only.

42. What is Special VFR, and when might you use it? **A:** ATC-authorized VFR flight within the surface area of Class B, C, D, or E airspace when weather is below basic VFR minimums but the pilot can remain clear of clouds with at least 1 SM visibility during the day. Must be requested by the pilot; at night, both pilot and aircraft must be instrument-current/equipped.

7. Cross-Country Flight Planning

43. What do you consider when planning a commercial cross-country flight — for example, one involving passengers or cargo on a schedule? **A:** The same core factors as any VFR flight planning — airspace, terrain/obstacles, checkpoints, weather, and alternates — plus, at the commercial level, weight and balance for the actual load being carried (passengers, equipment, cargo), fuel stops appropriate to that load, and a legal structure for the flight that avoids inadvertently crossing into common carriage.

44. How do you select a cruising altitude for a VFR cross-country? **A:** Balance terrain/obstacle clearance, winds aloft, and the VFR hemispheric cruising altitude rule above 3,000 ft AGL — odd thousands + 500 ft on magnetic courses 0°–179°, even thousands + 500 ft on 180°–359°.
45. What are the required elements of a VFR flight plan, and are you required to file one? **A:** Aircraft identification/type, true airspeed, departure point and time, cruising altitude, route, destination and estimated time en route, fuel on board, and number of persons on board. Filing is voluntary within the US — it exists for search-and-rescue protection, since Flight Service starts a search if you don't close it within about 30 minutes of your estimated arrival.
46. What fuel reserve is required for a VFR flight? **A:** Enough to fly to the first point of intended landing and, at normal cruise, continue for at least 30 minutes during the day or 45 minutes at night.
47. What minimum safe altitudes apply over open or sparsely populated areas, and why would this matter more to a commercial pilot than a typical private flight? **A:** Over anything other than a congested area, you may not operate closer than 500 feet to any person, vessel, vehicle, or structure. Commercial operations — aerial photography, survey work, low-altitude sightseeing — are far more likely to actually approach that limit than typical private cross-country flying, so knowing it cold (and building in your own margin, including gliding-distance-to-land considerations over water) matters more in practice.

8. Aerodynamics & Performance

48. Explain the two competing theories of how a wing generates lift. **A:** Bernoulli's principle — as air meets a restriction like the wing's curved upper surface, it speeds up and pressure drops, creating a pressure differential. Newton's third law — the wing deflects air downward and rearward; the equal-and-opposite reaction pushes the wing up. Both act together to produce lift.
49. What is load factor, and how does it relate to stall speed? **A:** The ratio of the load the wings are producing to the aircraft's actual weight, in G's. Load factor increases with bank angle in a level turn (2G at 60°) and increases stall speed by the square root of the load factor — a 2G load increases stall speed by about 41%.
50. What's VA (maneuvering speed), why does it change with weight, and what doesn't it protect against? **A:** The maximum speed at which a single, full, abrupt control deflection will stall the wing before exceeding the airplane's structural load limit. It decreases as weight decreases, because a lighter airplane stalls at a lower speed and sheds excess load sooner. It does not protect against multiple full-deflection inputs in sequence (full aileron immediately followed by full opposite rudder), which can still overstress the aircraft even below VA.
51. How does center-of-gravity location affect stability and efficiency, and why? **A:** A more aft CG reduces the arm between the CG and the tail control surfaces, making the aircraft less stable longitudinally — but it also requires less tail-down force to trim for level flight, which means a lower angle of attack, less induced drag, and better cruise efficiency. Forward CG trades the opposite way: more stable and generally more stall/spin-resistant, but less efficient.

52. What's the difference between indicated and calibrated airspeed, and where would you find the correction between them? **A:** Indicated airspeed (IAS) is simply what the instrument reads. Calibrated airspeed (CAS) corrects IAS for instrument and installation position error, which varies by airplane and even by configuration. The correction table showing the relationship between the two for a specific airplane is published in its POH.

53. What is wake turbulence, which aircraft generate the most, and how do you avoid it? **A:** Counter-rotating vortices off the wingtips of any lift-producing aircraft. The worst wake comes from aircraft that are heavy, clean, and slow. Avoid it by rotating before, and landing beyond, the point where a larger preceding aircraft rotated or touched down, staying above and upwind of its flight path.

9. Slow Flight, Stalls & Spins

54. What factors affect the airspeed at which a wing stalls? **A:** Angle of attack is the only direct cause of a stall, but weight, load factor, center of gravity, and power setting all change the airspeed at which critical AOA is reached in a given configuration.

55. What's an accelerated stall, and how does it differ from a normal power-off or power-on stall? **A:** An accelerated stall happens at a higher-than-normal airspeed because increased load factor (from a steep turn, or an abrupt pull) raises the stall speed above what it would be at 1G — a reminder that critical angle of attack, not a fixed airspeed number, is what actually triggers any stall. It can break more abruptly and with less warning than a 1G stall, since the margin between cruise speed and the elevated stall speed can be much smaller.

56. What's the proper stall recovery procedure? **A:** At the first indication: reduce angle of attack, add maximum available power smoothly, level the wings with coordinated aileron and rudder, and minimize altitude loss. Don't try to pick up a dropped wing with aileron alone — use rudder.

57. What causes a spin, and what's the aerodynamic difference between the incipient and fully developed phases? **A:** A spin begins with an uncoordinated stall — one wing more stalled than the other. In the incipient phase, the airplane is essentially rolling toward the more-stalled wing. As it develops, the yaw and roll couple together — aerodynamic forces from the rotation itself pin the rudder in a way that sustains and deepens the spin, which is why a developed spin needs a deliberate recovery input rather than simply neutralizing the controls.

58. What's the standard spin recovery procedure, and how do you identify which rudder to press if visual references are unavailable? **A:** PARE — Power to idle, Ailerons neutral, Rudder full opposite the spin direction, Elevator briskly forward to break the stall — then neutralize and recover from the dive (follow your specific POH if it differs). If you can see outside, you know your direction of rotation and press opposite it directly. Without visual reference, press against whichever rudder pedal is harder to move — the one being aerodynamically "pinned" by the spin is the one fighting you, and it's the wrong one.

10. High-Altitude Operations

59. At what altitudes is supplemental oxygen required for flight crew and passengers? **A:** Required crew must use it above 12,500 ft MSL for that portion of the flight lasting more than 30 minutes, and continuously above 14,000 ft MSL. Above 15,000 ft MSL, every occupant must be provided (not just offered) supplemental oxygen.

60. What is "time of useful consciousness," and why does it matter at high altitude? **A:** The amount of time a person can still function effectively after being deprived of adequate oxygen — it shrinks dramatically with altitude (many minutes in the teens of thousands of feet, down to under a minute above the flight levels most GA aircraft can reach). It matters because it's the actual window available to recognize a problem (a failed oxygen system, a pressurization loss) and take corrective action before impairment makes that impossible.

61. What types of supplemental oxygen delivery systems exist, and how do they differ? **A:** Continuous-flow systems deliver a steady flow regardless of breathing cycle, simple but less efficient. Demand systems deliver oxygen only during inhalation, more efficient. Pressure-demand systems deliver oxygen under positive pressure even during exhalation, used at the highest altitudes where continuous or plain demand systems can't maintain adequate blood oxygenation.

62. What precautions apply when using supplemental oxygen equipment? **A:** Keep it free of oil, grease, and petroleum-based products (a fire/combustion hazard in an oxygen-rich environment), don't smoke or permit an open flame near it, secure compressed gas cylinders properly against damage in turbulence or a hard landing, and use only aviation-grade "aviator's breathing oxygen" — not medical or welding oxygen, which can carry moisture that freezes and blocks the system at altitude.

63. What is cabin pressurization, and what physiological risk does a rapid decompression create? **A:** Pressurization maintains a cabin altitude lower than the aircraft's actual flight altitude by controlling the rate air is allowed to leave the cabin. A rapid decompression suddenly exposes occupants to the actual (much higher) outside altitude, sharply cutting time of useful consciousness and risking hypoxia before oxygen masks can be donned — the reason pressurized aircraft carry emergency oxygen with an immediate-deployment system rather than relying on stowed portable bottles alone.

11. Human Factors & Aeronautical Decision-Making

64. What is hypoxia — give the types, symptoms, and corrective action. **A:** Insufficient oxygen to body tissue, particularly the brain. Four types: hypoxic (not enough oxygen in the air), hypemic (blood can't carry oxygen — anemia, CO poisoning), stagnant (poor circulation — cold, high G-forces), histotoxic (cells can't use available oxygen — alcohol or drugs). Symptoms: euphoria, poor judgment, headache, drowsiness, tunnel vision, bluish lips/nail beds, eventual unconsciousness — often without the person recognizing the impairment. Corrective action: descend, use supplemental oxygen, treat the underlying cause.

65. A passenger becomes pale, can't catch their breath, and seems increasingly distressed at low altitude. How do you work through the possible causes? **A:** At low altitude, hypoxia from cabin altitude is unlikely, which narrows the field. Check for carbon monoxide first if there's any possibility of an exhaust leak — it's the most dangerous possibility — by considering recent preflight findings and whether cabin heat is in use. If CO is reasonably ruled out, the two most likely remaining explanations are hyperventilation (stress- or

anxiety-driven) and motion sickness, which can be worked through by asking how the person feels and what they were doing right before symptoms started (for example, repeatedly shifting focus between something close, like a camera screen or a chart, and the outside view is a common motion-sickness trigger).

66. What helps a passenger experiencing motion sickness? **A:** Point fresh air directly at them, have them focus on a fixed point on the outside horizon rather than anything up close, and avoid repeated near/far focus shifts (a camera or phone screen alternating with looking outside is a common trigger). If practical and safe, slow down and reduce maneuvering. Keep a sealable bag on hand, and don't allow anything to be thrown from the aircraft — improperly discarding an object, including vomit, over people or property below is treated as unsafe operation of an aircraft.

67. What causes carbon monoxide poisoning, and how do you guard against it? **A:** CO is a colorless, odorless gas from engine exhaust that can leak into the cabin — commonly through a cracked heat exchanger feeding the cabin heater — and binds to hemoglobin more readily than oxygen, causing a form of hypemic hypoxia. Symptoms (headache, dizziness, confusion, drowsiness) are easily mistaken for fatigue. Response: cabin heat off, fresh air vents open, land as soon as practical; carry and heed a CO detector, and check the exhaust system specifically during preflight if the aircraft's history or maintenance status is unfamiliar to you.

68. What are the hazardous attitudes, and what's the antidote to each? **A:** Anti-authority ("don't tell me") — follow the rules, they're usually right. Impulsivity ("do it now") — not so fast, think first. Invulnerability ("it won't happen to me") — it could happen to me. Macho ("I can do it") — taking chances is foolish. Resignation ("what's the use") — I'm not helpless, I can make a difference.

69. What is ADM, and what are SRM and CRM? **A:** Aeronautical Decision-Making is a systematic, continuous process for identifying hazards, assessing risk, and choosing the best course of action given the time and resources available. CRM is the use of all available resources by a multi-pilot crew; SRM is the same concept applied by a single pilot managing all internal (aircraft, personal condition) and external (ATC, weather, passengers) resources alone.

70. Why do personal minimums carry extra weight for a commercial pilot compared to a private one? **A:** A commercial pilot is more likely to be flying under external pressure from a client, employer, or paying passenger who has an interest in the flight happening regardless of conditions — the same "get-there-itis" risk that drives private weather accidents, but with a financial or professional relationship adding pressure on top. Setting and communicating personal minimums in advance, before a specific flight is on the line, is the defense against that pressure actually changing the decision in the moment.

12. Emergency Procedures

71. Engine fails at low altitude shortly after takeoff. What do you do? **A:** Pitch immediately for best glide and land essentially straight ahead, with only minor turns to avoid an obstacle. Below a pre-briefed altitude there is essentially never enough height to turn back to the runway — attempting the "impossible turn" is one of the classic fatal-accident scenarios in GA, and the altitude threshold below which you commit to landing straight ahead should be briefed before every takeoff.

72. Engine fails at cruise altitude with plenty of altitude available. Walk through your priorities. **A:** Fly the airplane first — pitch immediately for best glide and pick a suitable landing area considering wind, terrain, and obstacles. Only once that's established do you run the restart checklist (fuel selector/tank, mixture, carb heat/fuel pump, primer, mags), checking engine gauges (oil pressure/temperature) along the way for a diagnosable cause. If restart fails, secure the engine if time allows, declare an emergency if time/altitude allow, and brief passengers before configuring for landing.

73. What would you do for a gradual, unexplained loss of engine power and rising oil temperature during sustained low-altitude, low-airspeed, high-power flight — for example, during aerial work like photography or survey circling? **A:** Recognize this as a likely engine-overheating scenario — low airspeed and high power reduce cooling airflow over the engine. Clean up the airplane's configuration and increase airspeed to restore airflow, break off the low-and-slow work pattern, and monitor oil temperature and pressure closely; if either moves toward the red, treat it as a developing emergency and plan for a precautionary or forced landing rather than continuing the mission.

74. How does an in-flight engine fire response differ from an electrical fire? **A:** For an engine fire: mixture to idle cutoff, fuel selector/pump off to starve it, cabin heat and air off (to avoid drawing smoke into the cabin), and turn so the fire-side wing is into the wind to help blow it out, before transitioning to best glide and a forced landing. Don't attempt a restart once secured. For an electrical fire: master switch off first, then address individual equipment — you don't want to cut fuel to an engine fire that doesn't exist.

75. What's your responsibility, and ATC's role, if you declare an emergency? **A:** As PIC, in an emergency you have the authority to deviate from any regulation to the extent required to meet that emergency, though you may be asked to file a report afterward. ATC can provide priority handling, vectors, and coordination with emergency services — declaring early gives them more time to help.

76. What are your notification and reporting responsibilities after an accident or serious incident? **A:** Notify the NTSB "immediately" after an accident or a specific serious incident (flight control malfunction, in-flight fire), and file a written report within 10 days of an accident. Preserve the wreckage and records until the NTSB releases them, except as needed to protect life/property or in cooperation with law enforcement.

77. What emergency equipment is required, and what should a commercial pilot carrying passengers or cargo carry beyond the minimum? **A:** Required: an ELT, with limited exceptions. Beyond that, a fire extinguisher and first aid kit are strongly advisable for any passenger-carrying operation even when not federally mandated. Consider a handheld radio and portable GPS as backups, and gear suited to the specific mission — flotation and signaling equipment for extended overwater work, cold-weather survival gear for remote or mountainous routes.

13. Airport Operations & Communications

78. What ATC light signals might you receive if you lose radio communication in the pattern at a towered airport? **A:** Steady green: cleared for takeoff (ground) / cleared to land (flight). Flashing green: cleared to taxi (ground) / return for landing (flight). Steady red: stop (ground) / give way, continue circling (flight). Flashing red: taxi clear of runway (ground) / airport unsafe, do not land (flight). Alternating red/green: exercise extreme caution either way.

79. At a nontowered airport, how do you determine the active runway and pattern, and what's the recommended entry? **A:** Check the segmented circle and wind sock/tee at the field, and the Chart Supplement for the standard pattern altitude and direction (left traffic unless otherwise noted). The recommended entry is 45° to the downwind leg, at pattern altitude, entering at the midpoint of the downwind.
80. What are the basic right-of-way and communication procedures at a nontowered field? **A:** Self-announce position and intentions on CTAF — no clearance is issued, pilots see-and-avoid and sequence themselves. An aircraft established on final generally has priority regardless of how it got there, but that priority never permits cutting off another aircraft closer to the runway.
81. Describe the runway lighting and marking systems you might encounter. **A:** Runway edge lights are white (yellow for the last 2,000 ft on instrument runways). Threshold lights are green on approach, red from the runway. A VASI (2-bar) shows red over white for on-glidepath, white over white for high, red over red for low. A PAPI shows 2 red/2 white for on-glidepath, more white for high, more red for low.

14. Navigation

82. What is the difference between pilotage and dead reckoning? **A:** Pilotage is navigating by visually matching ground landmarks to the chart. Dead reckoning is navigating by calculation — heading, groundspeed, and time from true course, wind correction, and true airspeed — without reference to landmarks. VFR cross-country navigation normally combines both.
83. What magnetic compass errors should you understand? **A:** Variation is the fixed angular difference between true and magnetic north at a given location. Deviation is compass card error from the aircraft's own magnetic/electrical fields, unique to each aircraft. Dip errors occur during maneuvering — on north/south headings the compass leads or lags in turns; on east/west headings, acceleration or deceleration alone shows a turn that isn't happening.
84. What are the advantages and limitations of GPS versus VOR, and what is RAIM? **A:** GPS gives continuous, accurate position and direct navigation anywhere, but depends on satellite signal integrity and database currency. VOR is a simpler, independent, ground-based backup, but only defines specific radials from fixed stations. RAIM is a GPS receiver's internal check that enough satellites are in view and well-positioned to detect a faulty signal — a RAIM warning means the position can't be trusted and should be cross-checked another way.
85. How do you divert to an alternate airport in flight? **A:** Identify a suitable alternate on the chart, turn toward it promptly, and estimate heading and distance using the chart or a simple rule of thumb rather than exact calculation while also flying the airplane. Update ATC or Flight Following if in contact.
86. What do you do if you become lost or disoriented in flight? **A:** The "5 C's": Climb (for better visual range and radio/GPS reception), Communicate (contact ATC or Flight Service), Confess you're unsure of position, Comply with instructions given, and Conserve fuel. Maintain heading rather than wandering further, and look for large, distinctive landmarks to correlate with the chart.

15. Postflight Procedures

87. What do you do after landing and clearing the runway? **A:** Complete the after-landing checklist only once clear of the runway, behind the hold-short line — never while still on the runway. Taxi to parking per ground instructions, then complete the engine shutdown and securing checklist.
88. What must be logged after a flight, and what's the pilot's postflight responsibility? **A:** Flight time, landings (including night, for currency), and any discrepancies noted in flight for maintenance follow-up. The PIC remains responsible for leaving the aircraft in a safe, secure condition appropriate to the parking location and expected weather.
89. What should you do if you notice new damage or a discrepancy while securing the aircraft? **A:** Note it in the discrepancy log or notify the owner/operator/maintenance directly, and if it affects airworthiness, ensure the aircraft isn't flown again until a mechanic evaluates it — the same PIC airworthiness responsibility applies at the end of a flight as at the beginning.